

CAPSTONE I • OPEN TOPIC PRESENTATION

The Rise of AI Inference Chips

How specialized silicon is reshaping the cost, speed, and energy economics of AI

Presented by **Baps**

CSE423 — Capstone I | Summer 2026



Why AI Inference Chips?

Training was the spotlight. Inference is the bill.

Once a model is deployed, every user query costs compute. As LLMs scale, **inference now dominates real-world AI workloads** — and the economics of \$/token, watts/token, and tokens/second are reshaping the chip industry.

Why I chose this topic:

- It sits at the intersection of CS, hardware, and economics
- Hot industry shift happening right now — not five years away
- Directly impacts the cost and reach of every AI product I'll build

70%+

of AI compute spend
is now inference

10×

lower \$/token
from specialized chips

\$1T

projected AI hardware
TAM by 2030

Advantages — Why Inference Chips Win

1

Throughput

Purpose-built dataflow architectures deliver dramatically higher tokens/second than general-purpose GPUs.

2

Cost per Token

Specialized chips reduce \$/M tokens by an order of magnitude, making AI products economically viable at scale.

3

Energy Efficiency

Higher performance per watt — critical as data center power becomes the binding constraint on AI growth.

4

Lower Latency

On-chip SRAM and direct dataflow paths eliminate the memory-bandwidth bottleneck that throttles GPU inference.

Bottom line: *Specialization beats general-purpose silicon when the workload is well-defined and high-volume.*

Challenges — Why GPUs Still Dominate

1	The CUDA Moat	NVIDIA's software ecosystem (CUDA, cuDNN, TensorRT) is mature, ubiquitous, and hard to displace.
2	Memory Constraints	Many inference ASICs use small on-chip SRAM — large models require dozens of chips wired together.
3	Architectural Risk	Transformer-specific ASICs are a bet that the architecture won't change. If it does, the silicon is obsolete.
4	Ecosystem Maturity	Toolchains, debuggers, and frameworks lag behind NVIDIA — slowing developer adoption.
5	Training ≠ Inference	Most specialized chips can't train models, forcing a multi-vendor pipeline and operational complexity.

Bottom line: Hardware advantage is necessary but not sufficient — software and ecosystem decide who wins.

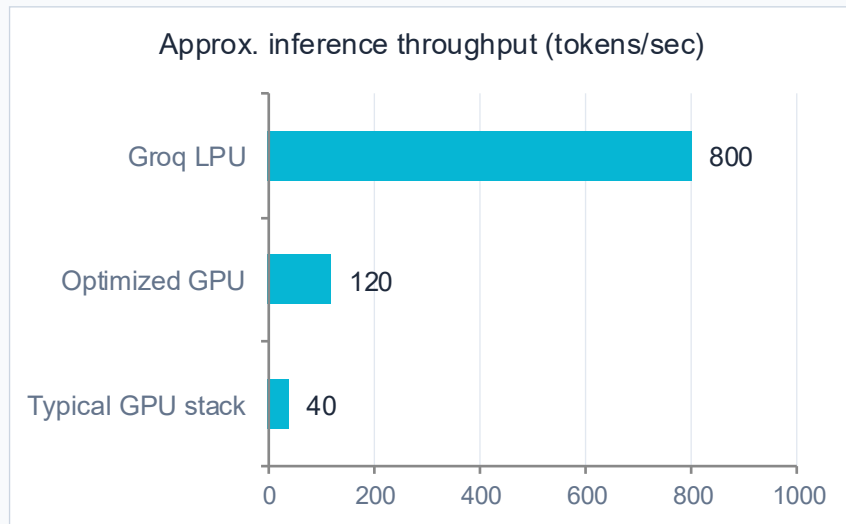
Case Study — Groq's LPU

What it is: A Language Processing Unit — a deterministic, dataflow-based inference chip designed specifically for transformer models.

Why it matters: Groq publicly demonstrated Llama-class models running at speeds that traditional GPU stacks could not match — proving the inference-specialization thesis with concrete numbers.

Key trade-off:

Each LPU has only 230 MB of on-chip SRAM — running a full LLM requires hundreds of chips networked together at the rack level.



Approximate published benchmarks for illustrative comparison.

~10× faster end-to-end response vs. typical cloud GPU inference

Future Potential

EDGE

On-Device Inference

NPU in laptops and phones (Apple Neural Engine, Qualcomm Hexagon) bring LLMs to the edge — private, instant, offline.

OPTICS

Photonic Computing

Companies like Lightmatter use light instead of electrons — promising orders-of-magnitude gains in energy efficiency.

MEMORY

In-Memory Compute

Analog and 3D-stacked memory architectures collapse the memory wall that limits today's GPUs and ASICs.

ASIC

Transformer ASICs

Etched, d-Matrix, and others are betting on transformer-only silicon — a high-risk, high-reward architectural wager.

Trend: AI hardware is fragmenting into purpose-built silicon for every layer — cloud, edge, and on-device.

Inference is the new battlefield.

01

The economics shifted

Inference now dominates AI compute spend — and small efficiency wins translate into massive dollar savings.

02

Specialization is winning

Purpose-built chips like Groq's LPU and AWS Inferentia prove the case for domain-specific silicon.

03

Software is the next moat

Hardware advantage alone won't beat NVIDIA — toolchains, libraries, and developer ecosystems decide the winners.

Why it matters to me: this is the silicon stack that every AI product I'll build my career on will sit on top of.

References



1. Groq, Inc. "Groq LPU Inference Engine." <https://groq.com> (2024).
2. NVIDIA. "H100 / H200 / Blackwell Architecture Whitepapers." <https://www.nvidia.com> (2023–2024).
3. Cerebras Systems. "Wafer-Scale Engine and Inference Benchmarks." <https://cerebras.net> (2024).
4. Etched. "Sohu — The First Transformer ASIC." <https://www.etched.com> (2024).
5. Amazon Web Services. "AWS Inferentia2 and Trainium Documentation." <https://aws.amazon.com> (2024).
6. Lightmatter. "Photonic Computing for AI." <https://lightmatter.co> (2024).
7. Patterson, D. & Hennessy, J. Computer Architecture: A Quantitative Approach, 6th ed. Morgan Kaufmann, 2019.
8. SemiAnalysis. "AI Inference Economics and the Cost-Per-Token Curve." <https://semianalysis.com> (2024).
9. The Information & The Wall Street Journal. AI hardware market coverage, 2023–2025.

Note: Numerical figures presented earlier are approximations drawn from publicly available benchmarks and industry analyses for illustrative purposes.